PIMPRI CHINCHWAD EDUCATION TRUST'S

PIMPRI CHINCHWAD COLLEGE OF ENGINEERING

Department of Applied Science and Humanities.

Report of FA-2 activity

Subject: Statistical Data Analysis using R

Academic Year 2024-25

Semester I

Submitted by 1.

Submitted to :

Date:

Sign:

Project Title : “Road Accident Analysis and Prediction ”

Data used: Road accidents dataset India 2014

Details of the data :

Column Name	Description
Reference Number	Unique identifier for each accident.
Grid Ref: Easting	The eastward geographical coordinate (Easting) of the accident location (in meters).
Grid Ref: Northing	The northward geographical coordinate (Northing) of the accident location (in meters).
Number of Vehicles	The total number of vehicles involved in the accident.
Number of Casualties	The total number of casualties (injuries or fatalities) resulting from the accident.
Accident Date	The date on which the accident occurred (DD-MM-YYYY format).
Time (24hr)	The time of the accident, represented in a 24-hour format (HHMM).
1st Road Class	The classification of the road (e.g., motorway, slip road, roundabout).
Road Surface	The condition of the road surface at the time of the accident (e.g., wet or damp, dry).
Lighting Conditions	The lighting condition at the time of the accident (e.g., daylight, darkness with no street lighting).
Weather Conditions	The weather conditions at the time of the accident (e.g., raining, no high winds, foggy).

Casualty Class	The class of casualty (e.g., driver, passenger, cyclist).
Casualty Severity	The severity of the injury (e.g., slight, serious, fatal).
Sex of Casualty	The gender of the casualty (male or female).
Age of Casualty	The age of the casualty at the time of the accident.
Type of Vehicle	The type of vehicle involved in the accident (e.g., car, lorry, motorcycle).

Problem statement 1:

Multilinear Regression for Predicting Accident Frequency Based on Road and Environmental Factors

Intro (Purpose and Approach):

The purpose of this analysis is to predict the frequency of accidents based on various factors such as the number of vehicles, road surface conditions, light conditions, weather conditions, speed limits, time of day, and day of the week. The approach involves preprocessing the dataset to handle missing values and transform categorical variables, followed by grouping the data for aggregation. A linear regression model is used to predict the accident frequency. The model's performance is evaluated using metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), accuracy, and precision, with an emphasis on classifying accidents based on frequency thresholds.

Approach / Algorithm with Explanation:

1. **Data Preprocessing:**
    - Missing values in crucial columns such as `Number_of_Casualties`, `Time`, `Road_Surface_Conditions`, and others are removed.
    - Categorical variables like `Road_Surface_Conditions`, `Light_Conditions`, and `Weather_Conditions` are converted to factors.
    - Time-related features (`Time_Hour` and `Time_of_Day`) are extracted and categorized into day or night.

    - The `Day_of_Week` feature is created to categorize dates into weekdays or weekends.
2. **Grouping and Aggregation:**
    - The data is grouped by factors such as the number of vehicles, road surface conditions, light conditions, etc., to summarize accident frequency and total casualties.
3. **Modeling:**
    - A linear regression model is built to predict accident frequency based on the aforementioned features and their interactions. Interaction terms allow the model to capture complex relationships between variables, reflecting how combinations of factors affect accident frequency.
4. **Thresholding and Classification:**
    - A **threshold** value of 10 is set to classify the predicted and actual frequencies into two categories:
        - **Class 1** (high accident frequency) if the predicted or actual frequency is greater than or equal to 10.
        - **Class 0** (low accident frequency) if the predicted or actual frequency is less than 10.
    - This threshold helps convert the continuous accident frequency predictions into binary outcomes for easier evaluation of classification performance.
5. **Prediction and Evaluation:**
    - The model's predictive performance is evaluated by comparing the binary predicted accident frequency classes to the actual binary classes.
    - **Accuracy** and **Precision** are calculated using the confusion matrix components (True Positives, True Negatives, False Positives, False Negatives).
    - The **Accuracy** measures the proportion of correct predictions (both high and low accident frequencies).
    - The **Precision** measures how often the model correctly predicted high accident frequencies among all predicted high frequencies.

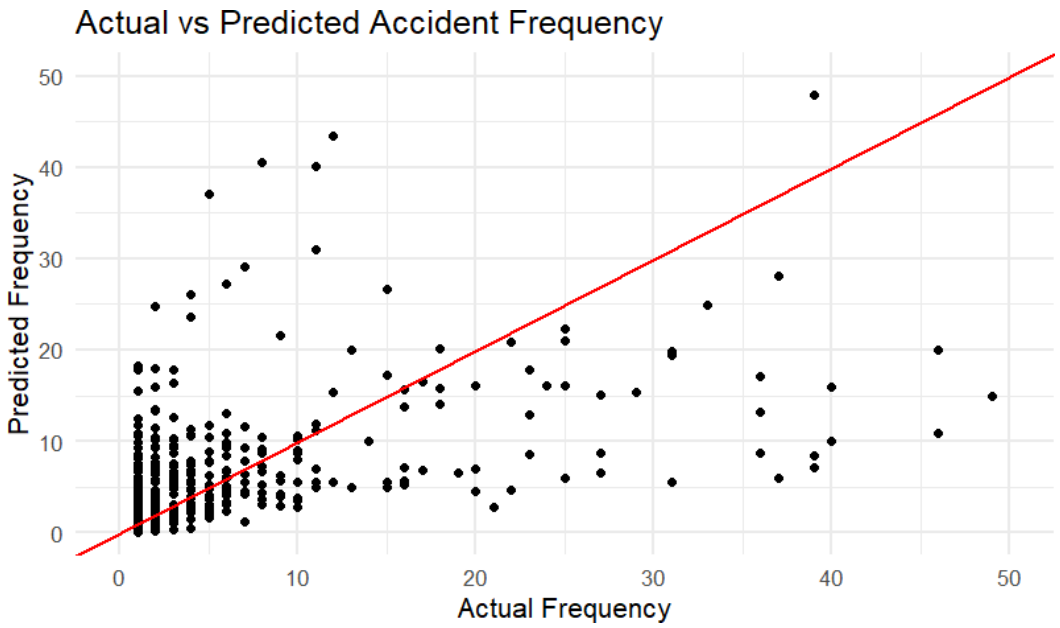
Output with Description:

The key outputs of this analysis include:

- **Model Summary:** The regression model produces an adjusted R-squared value of 0.6308, indicating a moderate fit.
- **Predictions:** The model predicts accident frequencies, which are then converted into binary classes based on the threshold (10).
- **Evaluation Metrics:**
    - **Accuracy:** 86.92%, indicating the proportion of correct predictions for both high and low accident frequencies.

- **Precision:** 64.29%, indicating the proportion of true positive predictions (high accident frequency) among all instances predicted as high accident frequency.

- **Visualization:** A scatter plot shows the relationship between actual and predicted frequencies, with a red line indicating perfect prediction.



Conclusion:

The inclusion of a threshold for classification improves the interpretability of the model by transforming the continuous accident frequency predictions into a binary classification problem (high vs. low frequency). The threshold of 10 is reasonable for identifying significant accident occurrences. The model demonstrates good accuracy (86.92%) in predicting accident frequency categories, although precision is lower (64.29%) and indicates that further improvements could be made in predicting true positives. Future work could explore using different thresholds or advanced models like decision trees or random forests to further refine predictions and improve the model's precision.

Problem Statement 2:

Impact of Time on Accident Frequency:

Analyze the distribution of accidents over different times of the day to identify peak hours for road incidents. Perform a hypothesis test to determine if accident frequency significantly differs during daylight versus nighttime hours, factoring in the lighting conditions.

Introduction:

This analysis aims to understand how time and lighting conditions impact road accident frequency. By examining the distribution of accidents throughout the day, the goal is to identify peak hours for road incidents. A hypothesis test will determine if accident frequencies differ significantly between daylight and nighttime, considering various lighting conditions. The purpose is to uncover patterns and differences in accident rates, which can inform targeted interventions to enhance safety measures.

Explanation:

Data Preprocessing:

1. Time data is cleaned and converted into numeric format for analysis.
2. The Lighting.Conditions variable is treated as a categorical factor (Daylight vs Nighttime).

Analysis of Accidents Over Time:

1. The dataset is grouped by the hour of the day to analyze accident frequency at different times.
2. A bar plot visualizes the accident distribution across different hours.
3. A scatter plot with a trend line is created to illustrate how accident counts vary by time.

Accident Proportions by Lighting Conditions:

1. The dataset is grouped by Lighting.Conditions (Daylight and Nighttime).
2. A pie chart is created to show the proportion of accidents under each lighting condition, providing insight into accident rates for Daylight vs Nighttime.

Hypothesis Testing: Chi-Square Test:

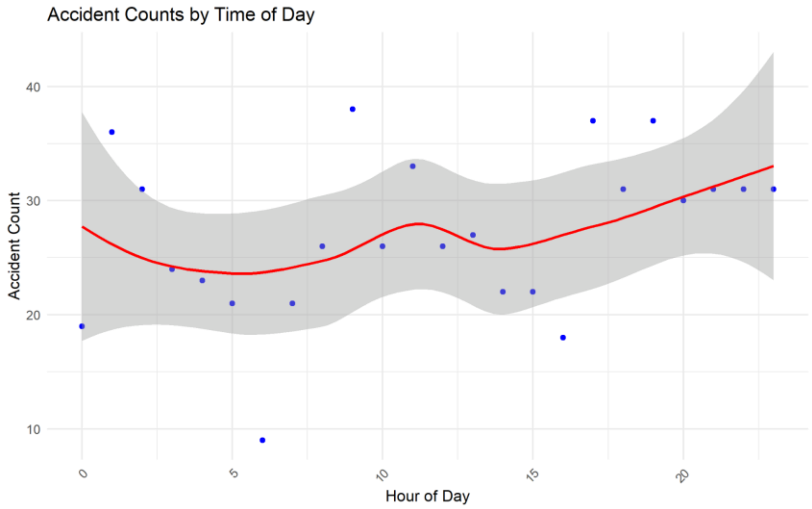
The **p-value** from the Chi-Square test indicates whether the difference in accident frequencies between Daylight and Nighttime is statistically significant. A p-value < 0.05 would suggest that accidents are significantly more frequent in one condition (e.g., Nighttime).

Bar Plot Comparison:

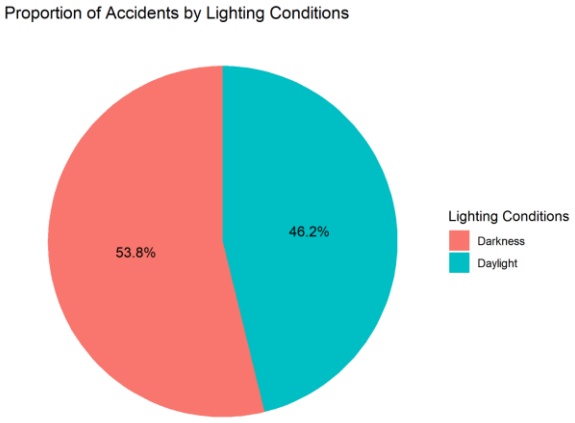
A bar plot compares the number of accidents between Daylight and Nighttime.

Output with description

- **Accident Counts by Time of Day:** This plot visualizes accident counts by hour of the day using blue points for each data entry and a red trend line to show the overall pattern. The x-axis represents the hour of day, while the y-axis shows the accident count.



- **Accident Proportions by Lighting Conditions:** This pie chart displays the proportion of accidents under different lighting conditions. Each segment represents a lighting condition, with its size proportional to the percentage of total accidents. The chart provides a clear visual comparison of accident frequencies across varying lighting scenarios.



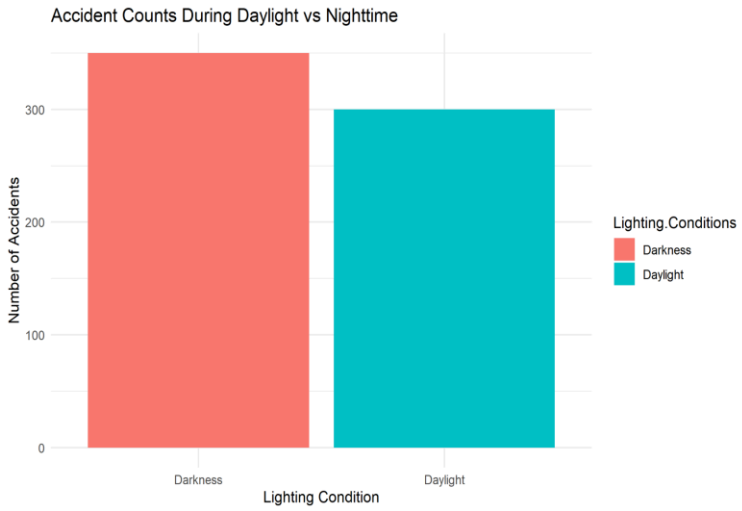
- **Hypothesis Test -Chi-Squared test - Daylight vs. Nighttime:**

  Chi-Square statistic (X Squared) : 7.3877.

  Degrees of Freedom (df): 1.

  p-value: 0.006567.

  The p-value of 0.006567 is less than 0.05, indicating a statistically significant difference in accident frequencies between Daylight and Nighttime. The test shows that lighting conditions likely affect accident rates.



Conclusion:

- The analysis reveals a statistically significant difference in accident frequencies between Daylight and Nighttime, with a p-value of 0.006567. The data suggests that accidents are more frequent during Nighttime, possibly due to reduced visibility and other factors. This highlights the impact of time of day on accident rates.

Problem statement 3:

Age-based Risk Analysis:

Investigate the relationship between the age of the casualties and accident severity.

Introduction:

The purpose of this analysis is to explore the relationship between the **age of casualties** and the **severity of accidents**. Specifically, we aim to determine whether there is a statistically significant association between the age group of individuals involved in accidents and the severity of those accidents. Understanding this relationship is important as it can highlight age-related risk factors in road safety, providing valuable insights for policymaking and the development of targeted safety interventions.

The approach taken involves analyzing accident data to examine how different age groups are associated with varying levels of accident severity. By conducting statistical tests, such as the **Chi-Square test of independence**, we will assess whether the distribution of accident severity differs across age groups. This analysis will help identify any patterns or trends, which can inform road safety campaigns and interventions tailored to specific age groups, such as younger or older drivers, that may be at higher risk for severe accidents.

Explanation

The following steps describe how the R script investigates the relationship between **age of casualties** and **accident severity**:

Step 1: Data Preparation and Sampling

In this step, the data is grouped by *Age_of_Casualty* and *Type_of_Vehicle*. A random sample of 100 observations per group is selected. This is done to ensure that the dataset remains manageable and provides sufficient data to perform the Chi-Square test.

Step 2: Contingency Table Creation

A contingency table (also known as a cross-tabulation) is a type of table used in statistics to summarize the relationship between two categorical variables. It displays the frequency distribution of the variables and helps determine if there is a significant association or dependence between them.

Structure of a Contingency Table

A contingency table organizes data into a matrix format, where:

- Rows represent the categories or levels of one variable.
- Columns represent the categories or levels of the second variable.
- Each cell in the table shows the frequency (or count) of occurrences of the combination of the corresponding row and column categories.

The table() function is used to create a contingency table that cross-tabulates the two variables of interest: Age_of_Casualty and Casualty_Severity.

Step 3: Chi-Square Test

The **Chi-Square test** is performed using the chisq.test() function on the contingency table created in the previous step. This function returns the **Chi-Square statistic**, **degrees of freedom**, and the **p-value**.
The p-value will help determine whether the relationship between age and accident severity is statistically significant. A small p-value (typically less than 0.05) suggests that there is a significant relationship between the two variables.

Output with Description:

After running the Chi-Square test, this is the obtained output:

```
        Pearson's Chi-squared test

data:  table_age_severity
X-squared = 16281, df = 188, p-value < 2.2e-16
```

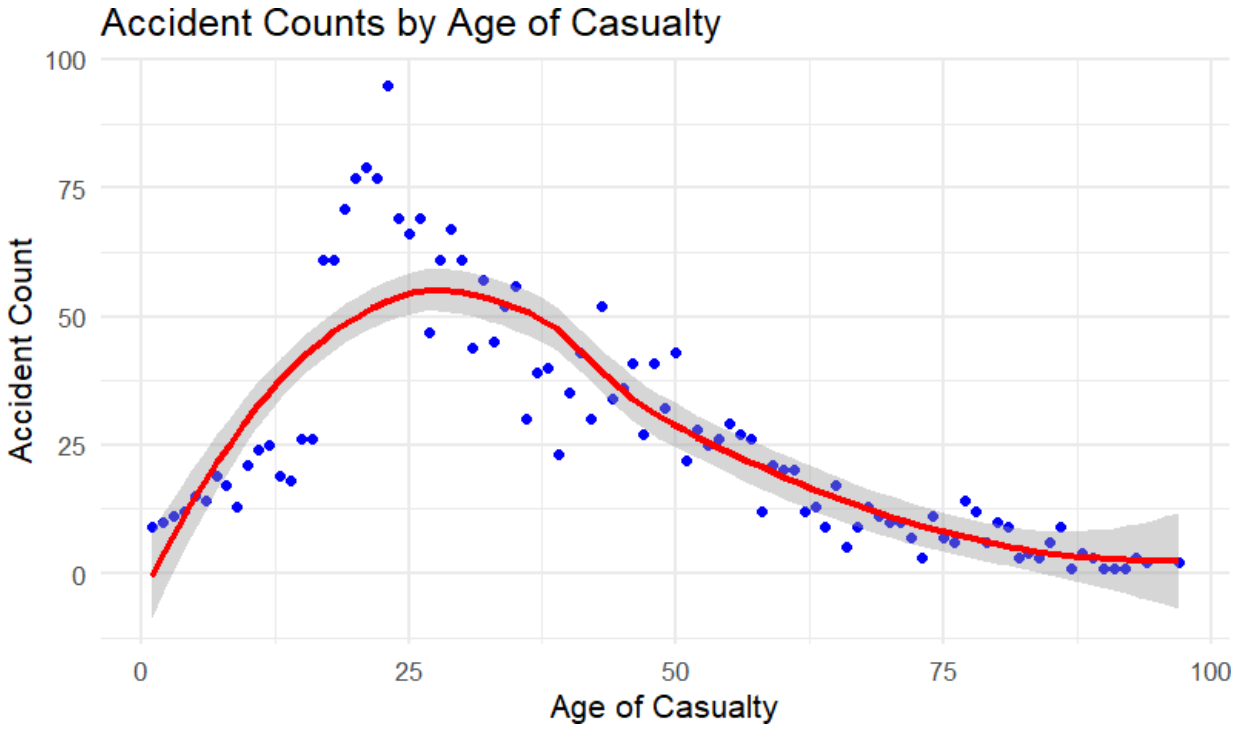
- **Chi-Squared Statistic**: The large value of the chi-squared statistic indicates a significant difference between the observed and expected frequencies of accident severity across different age groups.
- **Degrees of Freedom (df)**:

  This is calculated as (rows-1)*(col-1) for the contingency table.

  With 188 degrees of freedom, the test is considering a large number of possible comparisons, which strengthens the reliability of the results.

- **p-value**: The p-value is extremely small (<2.2×10−16< 2.2 \times 10^{-16}<2.2×10−16), much smaller than the conventional significance level of 0.05.

This suggests that the observed relationship between age and accident severity is **highly statistically significant**.



Conclusion : There is no similarity in Severity of Accidents across all the age groups,

Hence we can interpret that Severity of Accidents depends upon the Age of Casualty.

Problem statement 4:

Analyze the distribution of accidents over different days to identify peak days for road incidents. Perform a hypothesis test to determine if accident frequency significantly differs during weekdays versus weekends.

Explanation of the Problem Statement:

The problem aims to analyze road accident data to determine if there are specific days when accidents are more frequent. Specifically, it seeks to compare accident rates between weekdays and weekends and determine if the difference is statistically significant. This helps identify high-risk periods and inform traffic management strategies.

Purpose:

The aim of this analysis is to determine whether the frequency of road accidents significantly differs between weekdays and weekends. This information can help in resource allocation, traffic planning, and public awareness campaigns.

Approach:

1. Data Understanding and Preparation: Load and inspect the data to extract relevant columns, such as accident dates and the day of the week.
2. Visualization: Create bar plots to visualize the distribution of accidents across different days.
3. Hypothesis Testing:
    - Null Hypothesis (H_0): There is no significant difference in accident frequencies between weekdays and weekends.
    - Alternative Hypothesis (H_1): There is a significant difference in accident frequencies between weekdays and weekends.
4. Statistical Test: Perform a two-sample t-test or a non-parametric test (e.g., Mann-Whitney U test) to compare weekday and weekend accident frequencies.

Steps Performed:

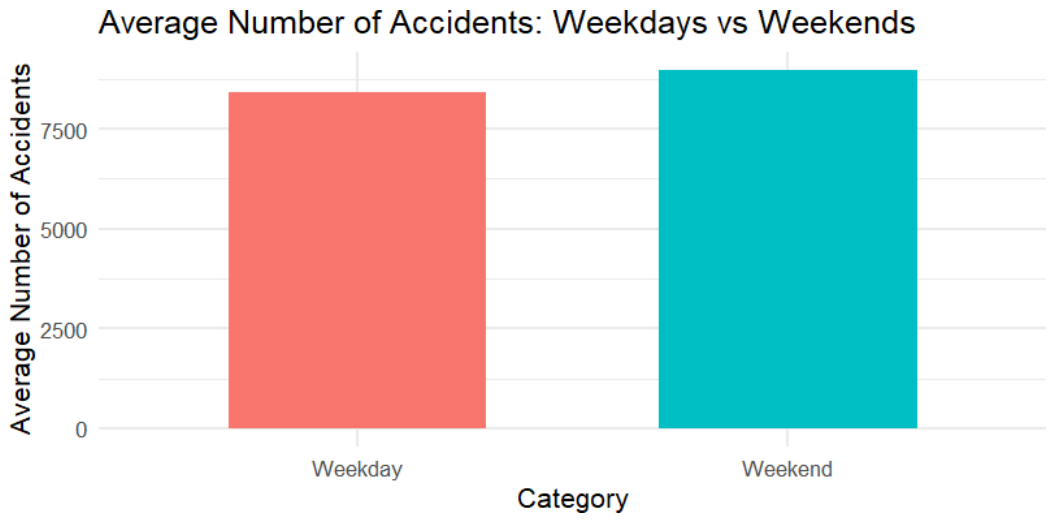
Steps:

1. **Load the dataset** to inspect its structure and check for necessary columns (e.g., accident date, day of the week).
2. **Aggregate the data** to calculate accident counts for each day.
3. **Visualize** the distribution of accidents across different days using bar plots.
4. **Separate weekdays and weekends** and perform a hypothesis test (e.g., a t-test or non-parametric test) to compare accident frequencies.
5. **Visualize** the comparison and provide a conclusion based on the test.

Output with Description

1. **Bar Plot of Accident Distribution**:

   This plot shows the number of accidents for days of the week, highlighting which days have higher incident rates.



2. **T-test Output**:

   The t-test compares the mean number of accidents on weekdays versus weekends and provides:

- **Hypothesis Test - Weekday vs. Weekend:**

  Test: Pearson's Chi-Squared test with Yates' correction.

  Chi-Square statistic (X Squared) : 15.594.

  Degrees of Freedom (df): 1.

  p-value: 7.851e-05.

  Proportion of Accidents During Weekday vs Weekend:

  Weekday: 0.0081 , Weekend: 0.0086"

Key Results:

- **Chi-Square Statistic (X^2)**: 15.594
  This indicates the magnitude of the difference between observed and expected frequencies.
- **Degrees of Freedom (df)**: 1
  This corresponds to the number of categories minus one for the day type (weekday and weekend).
- **p-value**: 7.851e-05 (or 0.00007851)
  The p-value is much smaller than the typical significance level of 0.05.

Interpretation:

- Since the **p-value < 0.05**, we **reject the null hypothesis (H_0)**.
- This indicates that there is a **statistically significant difference** in accident frequencies between weekdays and weekends.

Conclusion:

The analysis confirms that road accidents are not uniformly distributed across weekdays and weekends. Specific measures could be implemented based on these trends, such as heightened road safety enforcement or awareness campaigns during peak days.